

LongBee: Enabling Long-Range Cross-Technology Communication

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Abstract—Cross-Technology Communication (CTC) supports direct message exchange among heterogeneous wireless technologies (e.g., Wi-Fi, ZigBee, and Bluetooth) under the same ISM band, enabling explicit cross-technology control and coordination. For instance, a Wi-Fi AP can directly control ZigBee-enabled smart light bulbs without an expensive dual-radio gateway. Such CTC capability can be further amplified if we can extend the communication range of CTC to support long-range wide-area IoT applications such as environmental monitoring, smart metering, and precision agriculture. Our work, named LongBee, is the first to extend the range of Cross-Technology Communication. At the transmitter side, LongBee concentrates the effective TX power through down-clocked operations, and at the receiver side, LongBee improves the RX sensitivity with an innovative transition coding to ensure reliable preamble detection and payload reception. All these are achieved without modifying hardware and without introducing extra Wi-Fi RF energy cost.

We implemented the LongBee on the USRP platform and commodity ZigBee devices. Our comprehensive evaluation reveals that LongBee with concentrated TX power and higher RX sensitivity achieves reliably over 10x range extension over native ZigBee communication and 2x range extension than the longest distance achieved by existing CTC schemes so far.

I. INTRODUCTION

According to Gartner [9], the number of Internet of Things (IoT) devices connected wirelessly will reach 20 billion by 2020, supporting many promising IoT applications such as smart city [16], environmental monitoring [25], precision agriculture [11] and industry control [21].

To meet diverse application requirements, researchers utilize a wide range of wireless technologies to strike a balance among bandwidth, range, reliability, and cost. These technologies (e.g., Wi-Fi, ZigBee, and Bluetooth) often share the same unlicensed spectrum (e.g., ISM bands). Although such spectrum sharing would cause severe cross-technology-interference, it also offers a new opportunity for cross-technology communication (CTC). For example, a FreeBee Wi-Fi AP [14] can directly control ZigBee-enabled home appliances without the assistance of costly (> 150 USD) dual-radio gateways.

Recent advances, WEBee [15] in particular, open a new frontier for direct communication between heterogeneous devices by implementing a physical-layer cross-technology communication channel. In a nutshell, by constructing a selected Wi-Fi payload, a physical-layer CTC (e.g., WEBee) sends a ZigBee-compliant frame from a commercial Wi-Fi card (e.g., Atheros AR2425) to commercial ZigBee devices (e.g.,

CC2420). Hypothetically, with a high transmission-power Wi-Fi sender (e.g., 20dBm) and a highly sensitive ZigBee receiver (e.g., -95dBm), one would expect naturally 10x times range extension (i.e., 20db gain) in comparison with that of native ZigBee-to-ZigBee communication. If the wireless communication range is extended, we can enjoy a few benefits. For examples, we can eliminate the hefty cost of multi-hop relays if one Wi-Fi AP can communicate with ZigBee devices directly over a large area (e.g., $1km^2$) in replacement of a multi-hop ZigBee mesh network. With a smaller number of hops, we can achieve better end-to-end reliability and reduce idle-listening energy at relay nodes significantly.

To test our hypothesis, we conduct a few field experiments using WEBee [15]. It turns out that the effective range extension is constrained by the intrinsic bandwidth asymmetry between Wi-Fi and ZigBee (20MHz vs. 2M) and signal distortion under imperfect emulation. To address these constraints, this work introduces LongBee, a long range CTC scheme based on two innovative designs. First, LongBee concentrates Wi-Fi TX power by shrinking the transmission bandwidth of Wi-Fi through down-clocking operations; Second, LongBee develops a transition coding mechanism to combat the inherent defects of the emulated signals, which deteriorates the ZigBee receivers' sensitivity. With these two designs, LongBee can achieve highly reliable wireless communication between Wi-Fi and ZigBee over 300 meters in outdoor environments and over 100 meters for indoor environments. Such a range is over 2x of state-of-art CTC and about 10x of standard ZigBee communication with 0dBm transmission power. Specifically, the contributions of LongBee are:

- We design LongBee, a long range CTC scheme by manipulating only the payload of Wi-Fi frames without modifying the hardware and requiring no RF amplifier. In other words, such extension is achieved without additional hardware and extra RF energy cost.
- We address two important challenges, namely (i) concentrating the effective transmission power of CTC without increasing overall transmission power, and (ii) improving the receiver sensitivity of the emulated physical-layer signals without changing receivers' RF front end. These techniques provide general guidance for future long-range communication.
- We implement and evaluate LongBee on USRP-B210 and

commodity ZigBee devices (e.g., CC2420). Our extensive evaluation demonstrates that LongBee can transmit a ZigBee frame from a Wi-Fi sender to a ZigBee receiver under long-range settings, easily outperforming both native ZigBee communication and the state-of-the-art CTC designs.

The rest of the paper is organized as follows. Section II motivates the need for long range communication. Section III overviews the LongBee architecture. Sections IV and V describe the two key design elements of LongBee, i.e., the down-clocked emulation and transition coding mechanism. Section VI presents the testbed and evaluation of LongBee. Section VII discusses the related work, followed by the conclusion in Section VIII.

II. MOTIVATION

Internet of Things (IoT) is revolutionizing the way we live and work. These *things* will play an important role in many applications including but not limited to the smart city, intelligent transportation system, environment monitoring, smart metering, supply chain, industry control and precision agriculture. In many such applications, the IoT devices are sparsely deployed in a wide area. To provide connectivity, both multi-hop meshing networking and single-hop long range communication are proposed by researchers in the community. Although it is a subject of much debate, this section attempts to explain the limitation of mesh networking, serving as the motivation for long-range communication.

The Curse of ZigBee mesh networking: ZigBee is a typical type of short-range wireless technology. Constrained by its transmission power, ZigBee can only transmit 20~30 meters in realistic environments. Consequently, a very dense deployment is required to cover a larger area/distance, leading to hefty cost regarding hardware, energy, maintenance, and performance. For examples, UC Berkeley [13] constructs a 48-hop wireless network to cover the Golden Gate Bridge which spans 4,200 feet, achieving only an 800B/second low throughput to guarantee end-to-end reliability. To cover a 1,300 by 300 meters area, the ExScal project [2] deployed a 1000+ XSM nodes for military surveillance, revealing significant issues related to end-to-end routing yield, deployment overhead, localization faults and management costs. Admittedly short-range mesh networking is appropriate in some particular settings, however, from the practical point of view, long-range communication reduces unnecessary deployment of relay nodes and reduces the cost and the complexity of IoT networks dramatically. Therefore, it is more desirable in many cases.

The Blessing of Cross-Technology Communication: Inherently constrained by the hardware design and power constraints, the communication range of ZigBee devices are limited. Fortunately, a recent breakthrough in CTC [15] provides a new opportunity to increase the connectivity of low-power IoT devices (e.g., ZigBee) with the help of high-power wireless technologies (e.g., Wi-Fi and LTE). For example, the transmission power of Wi-Fi devices can be as high as 30dBm

(1000x more than that of ZigBee devices), allowing a Wi-Fi device sends RF signal detectable by the ZigBee devices from a much longer distance.

Different from other solutions such as using high-transmission power or special hardware (e.g., TinyNode[7], Fleck-3[6], and SNOW[18]), extending communication range with CTC comes with a few unique benefits: First, the range extension is achieved without additional hardware and energy cost. Second, it eliminates the need for costly gateways and customized hardware, and third, cross-technology communication itself can broaden the scope of IoT applications through cross-technology collaboration.

The Room for CTC Range Extension: Recently WEBee [15] implemented a physical-level cross-technology communication, in which a Wi-Fi device transmits a frame with a selected payload that can be recognized by a ZigBee receiver as a legitimate ZigBee frame. Because a Wi-Fi sender normally transmits at 20dBm, such signal can propagate a longer distance than that of ZigBee signal. For example, it has been shown WEBee [15] can extend the effective range of ZigBee communication to 100 meters. Although the result is encouraging, our recent study reveals that some inherent constraints of WEBee still severely limit the wireless communication distance. For example, a ZigBee node can only receive a small portion of TX energy emitted by the Wi-Fi sender, because a Wi-Fi sender occupies a much wider bandwidth (20MHz) than that of ZigBee (2MHz). These constraints motivate us for a better design as presented in the rest of the paper.

III. OVERVIEW OF THE LONGBEE DESIGN

Inspired by the work of WEBee, we design and implement a new CTC scheme, named as LongBee, with much-enhanced communication range. Specifically, LongBee combines the strengths of two heterogeneous technologies, i.e., the high transmission power of Wi-Fi and the fine receiver sensitivity of ZigBee, to increase the wireless communication range significantly. Technically, LongBee utilizes quarter-clocked Wi-Fi OFDM to emulate the ZigBee signal with concentrated TX power and provides a transition coding to improve the receiver sensitivity when demodulating the emulated ZigBee signal. We note although LongBee is a specific long-range CTC from Wi-Fi to ZigBee, the features and challenges we address in this paper are generic and applicable to other long-range low-power wireless systems based on CTCs. In the rest of this section, we start with preliminaries on wireless communication and briefly overview the design of WEBee and LongBee.

A. Range of Wireless Communication

The range of wireless communication is constrained by the radiation laws, i.e., the power-distance relationship characterized by a Friis free space equation [20]:

$$P_r = \frac{P_t G_t G_r \lambda^2}{(4\pi d)^2} \quad (1)$$

where P_r and P_t are the received and transmitted signal power, respectively, G_r and G_t denote the antenna gains at the receiver and transmitter, λ is the wavelength of the transmitted signal, and d is the communication distance corresponding to the LOS path.

When using “dBm” to restate the equation (1), we have a simplified relation as following

$$lg(d) \propto TxPower(dBm) - RxSensitivity(dBm), \quad (2)$$

As indicated by equation (2), to support long range wireless communication, one can either (i) increase the transmission power or (ii) improve the receiver sensitivity (or do both).

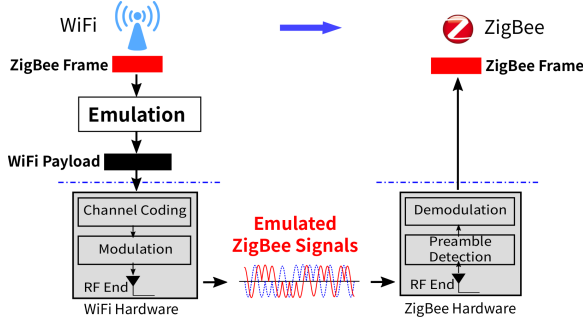


Fig. 1. How WEBe Works.

B. Background of WEBe

WEBe [15] is the first work that can emulate ZigBee frames with Wi-Fi payloads without any hardware and firmware modification. Fig. 1 illustrates how WEBe works: a Wi-Fi device transmits a frame with a *selected* payload, and the waveform corresponding to the Wi-Fi frame can be recognized by a ZigBee receiver as a *legitimate* ZigBee frame. More specifically, WEBe constructs the payload of a Wi-Fi frame elaborately so that the RF waveform resembles ZigBee signals. This waveform is synthesized automatically when the selected Wi-Fi payload goes through the channel coding and modulation of Wi-Fi transmitter. When the Wi-Fi waveform or the emulated ZigBee signal is transmitted into the air through the front end of Wi-Fi, it will successfully pass the ZigBee preamble detection and be demodulated at the ZigBee receiver as a legitimate ZigBee frame.

C. Overview of LongBee Design

In terms of signal emulation, LongBee works similar to WEBe. But different from WEBe, LongBee achieves a much longer communication range by concentrating the TX power and improving the RX sensitivity. Fig. 2 illustrates the whole procedure of the LongBee.

- In the first step, an original ZigBee frame is converted to a sequence of ZigBee symbols through our transition encoding mechanism. The purpose of encoding is to make emulation in the second step more precise, hence improving the RX sensitivity.
- In the second step, this encoded sequence is fed into a module named as down-clocked emulation to produce the corresponding Wi-Fi payload. The purpose of down

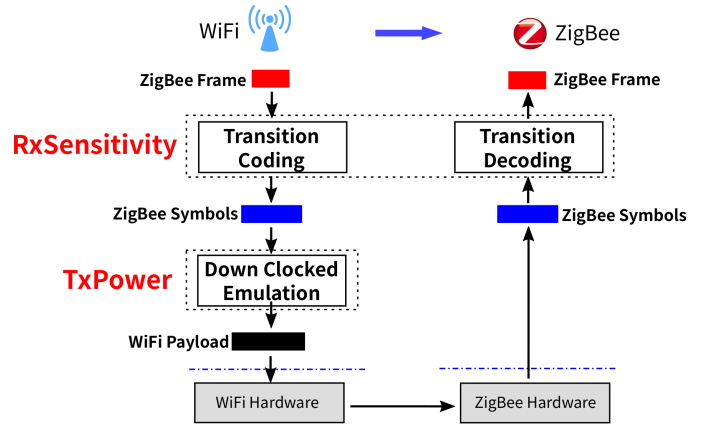


Fig. 2. How LongBee Works.

clocking is to concentrate Wi-Fi energy within ZigBee’s bandwidth, hence increasing the TX power.

- In the last step, this Wi-Fi payload is transmitted by the Wi-Fi hardware, the ZigBee hardware will demodulate and output the sequence of encoded ZigBee symbols. Finally, a module, named as transition decoder, extracts the original ZigBee frame and feeds it into the application layer. In the next two subsections, we explain in more details how the communication range is extended.

Design I: Boost the Effective TX Power with Down Clocking

The transmission power of Wi-Fi is much higher than that of ZigBee, but the bandwidth of Wi-Fi is also much wider in comparison with ZigBee. In Fourier transform analysis, the Parseval’s theorem[24] states an equation of energy in the time-domain and frequency-domain, i.e.,

$$\int_{t=-T/2}^{T/2} |u(t)|^2 dt = T \sum_k |U[k]|^2, \quad (3)$$

where $u(t)$ are time domain signals and $U[k]$ are corresponding DFTs, and the integral of the squared magnitude of a function is known as the energy of the function.

Parseval’s theorem shows that the transmission power of Wi-Fi will be distributed into the whole Wi-Fi bandwidth (20MHz). However, only partial transmission power in small part of Wi-Fi’s band overlapped with ZigBee (2MHz) is effective for ZigBee receivers. Therefore, the first major design of LongBee is to boost the effective transmission power of the Wi-Fi sender within ZigBee bandwidth.

According to IEEE 802.11-2007 standard[1], the OFDM-based PHY supports two down-clocked operations with 10MHz (half-clocked) and 5MHz (quarter-clocked) sampling-rate, in addition to the default full-clocked 20MHz operation. Recently, the IEEE 802.11ah standard [22] also supports flexible bandwidths, i.e., 2MHz, 4MHz, 8MHz, and 16MHz. To be compatible with the standard, down clocking is supported in many commercial Wi-Fi cards (e.g., Atheros 5414).

The bandwidth of Wi-Fi transmitted signals shrinks in the down-clocked operation mode, so the effective transmission power of the Wi-Fi in the ZigBee band increases significantly.

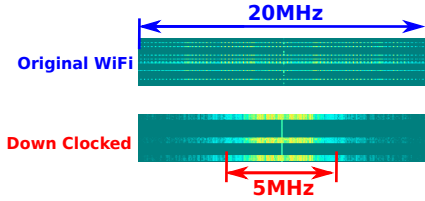


Fig. 3. Spectrum of Down-clocked Wi-Fi.

Fig. 3 illustrates the spectrum of received signals with different clocked mode but same transmission power. Fig. 3 shows that the effective received power at the ZigBee receivers is boosted when the Wi-Fi transmitter works in a quarter-clocked operation mode. Although the concept of down clocking is straightforward, it brings a few challenging issues for CTC as we present later in Section IV.

Design II: Improve the RX Sensitivity with Transition Coding: WEBe constructed the ZigBee signals without any change of commercial Wi-Fi hardware [15]. Due to hardware constraints in Wi-Fi such QAM quantization and cyclic prefixing, WEBe can't emulate ZigBee signals perfectly. If the more precisely Wi-Fi can emulate the ZigBee signals, the better chance a ZigBee receiver can decode the frame. In other words, the RX sensitivity can be improved through a better emulation.

The second major design of LongBee is to improve the receiver sensitivity by handling the inherent signal emulation defects in WEBe. Specifically, we found that a certain set of ZigBee signals could be emulated better under the Wi-Fi hardware constraints than a random sequence of ZigBee signals, and accordingly, we should code the ZigBee frame based on these "better emulated" signals before emulation to improve the receiver sensitivity of LongBee significantly.

IV. DOWN-CLOCKED EMULATION

We note that the maximum transmitter output power allowed by FCC under ISM band is 30dBm [8]. Therefore it is not legitimate to achieve a longer communication range by simply amplifying the TX signal blindly. In this section, we explain how to concentrate transmission energy at the Wi-Fi transmitter through down clocking and addressing related emulation challenges.

A. Channel Mapping

Wi-Fi OFDM (Orthogonal Frequency Division Multiplexing) system provides a "quarter-clocked" operation in which Wi-Fi uses 5 MHz channel spacing. We also note that the guarding band of each ZigBee channel is also 5MHz. With such "quarter-clocked" operation, one ZigBee band is precisely embedded into one Wi-Fi band, resulting in the least waste of transmission power. The central frequency of Wi-Fi and ZigBee are not aligned precisely, therefore we need to align them. We note many commodity Wi-Fi radios (e.g., Atheros AR9485, AR5112, and AR2425) and ZigBee devices (cc2420, cc2530) can set their central frequency, hence channel mapping can be easily achieved at the either Wi-Fi or ZigBee

side. Fig. 4 illustrates the channel mapping with such Wi-Fi operation, i.e., the channel mapping used by LongBee:

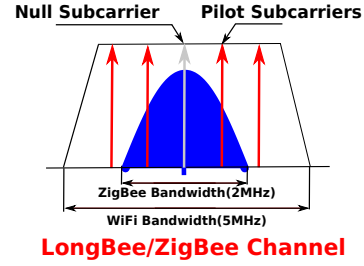


Fig. 4. Channels Mapping for Down Clocked Emulation

B. Signal Emulation

It is known that Wi-Fi OFDM uses the frequency signals in subcarriers to represent (modulate) the payload. And Wi-Fi uses Inverse Fast Fourier Transform (IFFT) to transform the frequency signal into a time-domain signal and sends it into the air. Conversely, if we want to emulate a desired ZigBee time-domain signal, we feed the time signal into the FFT to obtain the corresponding frequency components.

The quarter-clocked operation quadruples symbol times by down-sampling. So in this operation mode, the duration of a Wi-Fi symbol becomes $16\mu s$ instead of $4\mu s$ of normal Wi-Fi mode. It is known that the duration of a ZigBee symbol is also $16\mu s$. So a desired ZigBee symbol is fed into the FFT to obtain the Wi-Fi frequency signals as shown in Fig. 5.

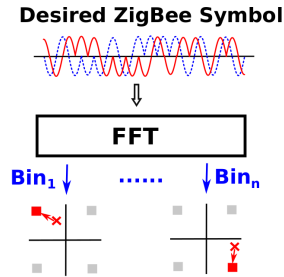


Fig. 5. Signal Emulation of LongBee.

Challenge I: Minimizing quantization error: Wi-Fi modulation can't represent arbitrary frequency signals. Instead, Wi-Fi uses predefined discrete QAM (Quadrature Amplitude Modulation) points to represent the frequency components. So the frequency components for each FFT bin (i.e., Wi-Fi subcarrier) in Fig. 5 need to be quantized and matched to the discrete QAM points.

The challenging issue is to achieve such quantization with minimal errors. In Fourier transform analysis, the Parseval's theorem[24] states an equation of energy in the time-domain and frequency-domain. Combined with the linear property of Fourier transform, we have the following equation to state the errors introduced by frequency-domain quantization, i.e.,

$$\int_{t=-T/2}^{T/2} |u(t) - \hat{u}(t)|^2 dt = T \sum_k |U[k] - \hat{U}[k]|^2, \quad (4)$$

where the $u(t)$ is the desired time domain signal and $\hat{u}(t)$ is the time domain signal after QAM quantization and $U[k]$ and $\hat{U}[k]$ are corresponding DFTs.

The equation (4) shows that minimizing the signal distortion in the time-domain under energy metric is equivalent to minimizing the total error of quantization in frequency components. Therefore, we choose the closest QAM points in term of total Euclidean distance to each k FFT points of desired signals as shown in Figure 5 to minimize the quantization error.

Challenge II: Dealing with Pilot and Null Subcarriers: Not all frequency components in FFT bins can be quantized into QAM points. The 5 MHz Wi-Fi OFDM is composed of not only data subcarriers but also null subcarriers and pilot subcarriers for channel state, sampling offset and frequency offset estimation. Fig. 4 shows that the FFT bins of the desired ZigBee signal overlap with two pilot subcarriers and one null subcarrier of Wi-Fi, which are out of our control.

We note that the pilot signal is predefined in Wi-Fi OFDM [1], which is either $(-1,0)$ or $(1,0)$ in the frequency domain plane. If quantization error caused by the pilot signal is as large as shown in the left part of Fig. 6, the emulation quality is reduced. Since we can not change frequency-domain pilot signal which is fixed by Wi-Fi, we have to select the right emulation target: the time-domain waveform of ZigBee signals. Technically, if we flip time domain signals, the frequency components of this signal are flipped as well, because $F[-X] = -F[X]$ holds for FFT transform F . Therefore, illustrated as Fig. 6, flipping the original desired ZigBee signals may produce a frequency component in the pilot bin which is much closer to the predefined Wi-Fi pilot point. On the other hand, ZigBee demodulates the signals by computing the phase shifts between two consecutive complex samples. The phase shift of samples $s(n)$ and $s(n-1)$ is computed by $\arctan(s(n) \times s^*(n-1))$, where $s^*(n-1)$ is the conjugate of $s(n-1)$. Obviously, $s(n) \times s^*(n-1) = -s(n) \times -s^*(n-1)$, so the flipping shown in Fig. 6 won't influence the ZigBee demodulation.

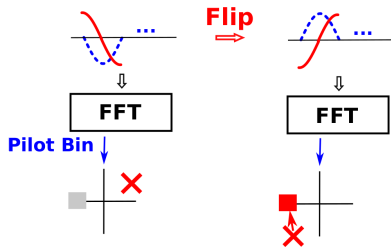


Fig. 6. Pilot Subcarriers and Signal Flipping.

For the null subcarrier shown in Fig. 4, it corresponds to the direct circuit of desired ZigBee signals. We note that the direct circuit of all ZigBee signals is zero due to the symmetry of ZigBee OQPSK modulation. Moreover, the flipping won't change the symmetry of the ZigBee signals and won't change the direct circuit also. In other words, null subcarrier will not cause emulation distortion at all and we can safely ignore it.

V. TRANSITION CODING

The inherent gap between Wi-Fi modulation and ZigBee demodulation brings out intrinsic errors in emulated signals, reducing the receiver's sensitivity. This section improves the demodulation sensitivity of ZigBee by mediating the impact of Wi-Fi Cyclic Prefixing (CP).

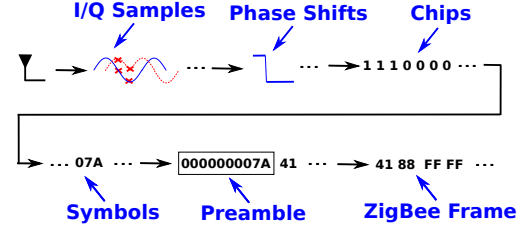


Fig. 7. Diagram of ZigBee Receiver

A. Background

Demodulation at the ZigBee Receiver: The physical layer of ZigBee receivers is shown in Figure 7. The ZigBee RX Radio firstly down-converts the passband waveforms to baseband and digitalizes them into in-phase/quadrature (I/Q) samples. The demodulation of ZigBee is Offset Quadrature Phase-Shift Keying (OQPSK), in which the phase shifts between the consecutive I/Q samples are computed. With a simple thresholding, ZigBee outputs the chip value “1” if the phase shift is bigger than 0° and otherwise outputs “0”. After collecting 32 chips, ZigBee maps these chips into four bits (i.e., a ZigBee symbol), according to a predefined symbol-to-chip table.

As shown in Figure 7, once the symbols have been decoded, a ZigBee receiver recognizes the physical-layer frame through a state machine. Specifically, a ZigBee receiver firstly identifies the preamble, i.e., a predefined sequence of symbols composed of hexadecimal value of 000000007A by default. If and only if the correct preamble has been identified, a ZigBee receiver goes into a recording state to demodulate following symbols and record them into a link-layer frame.

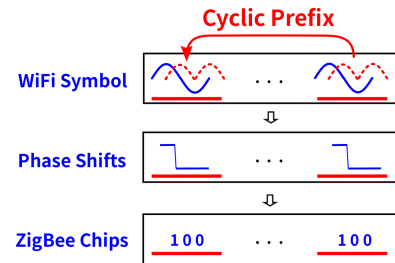


Fig. 8. Cyclic Prefix & ZigBee Demodulation

Cyclic Prefixing at the Wi-Fi transmitter: To combat the multi-path effect in complex environments, the Wi-Fi time-domain signal is processed by a module named as Cyclic Prefixing (CP). As shown in 8, Cyclic Prefixing copies the right part of Wi-Fi symbol and pasted into (overwrite) the left of the symbol. As a result, the front segment and the end segment of Wi-Fi symbol are the same. Obviously, such repetition by a Wi-Fi transmitter will produce repeated phase shifts and produces repeated chips when a ZigBee receiver

demodulates as shown in Fig. 8. Unfortunately, there is no such chip repetition in standard ZigBee symbols. Therefore, the Wi-Fi CP module will unavoidably cause several chip-errors for every ZigBee symbols, reducing the receiver's sensitivity. The transition code is designed to reduce such chip-error introduced by Cyclic Prefixing.

B. Transition Coding for Better Preamble Detection

It is noted that we can introduce redundancy through conventional coding schemes into the payload of ZigBee frame to improve the receiver sensitivity, but *we can not add redundancy in ZigBee preamble because the structure of preamble is fixed by ZigBee's hardware*. This unique challenge makes our transition coding an unprecedented design.

In order to understand transition coding, we start with an introduction of preamble detection. To achieve synchronization between a transmitter and a receiver, a preamble is prefixed for each frame transmission. Specifically, a ZigBee preamble consists of a sequence of zero symbols followed by a SYNCWORD. The detection of SYNCWORD indicates the exact starting time of the payload.

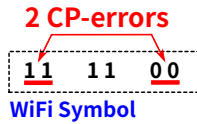


Fig. 9. Emulation Errors due to CP

Because preamble detection is implemented normally with a hardware correlation detector, we can not change the structure of the preamble. The only choice to improve the detection quality is to select the best ZigBee SYNCWORD for Wi-Fi emulation. For example, if the binary chip sequence of the ZigBee SYNCWORD symbol is “111100” as shown in Fig. 9¹, Wi-Fi Cyclic prefixing (i.e., prefixing is a repetition of the end) will always causes 2 chip errors.

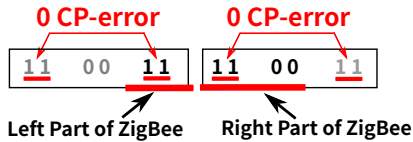


Fig. 10. Zero Emulation Error with Transition Coding

To address this issue, we introduce here transition coding. The key idea is to split a ZigBee SYNCWORD symbol into **left part** and **right part**, and use the first Wi-Fi symbol to emulate the left part of SYNCWORD and the second Wi-Fi symbol to emulate its right part. For example, as shown in Fig. 10, the SYNCWORD symbol “111100” is split into “11” and “1100”, and is perfectly emulated by two consecutive Wi-Fi symbols with chip sequence “110011”. Note the sequence “110011” has the same prefix and suffix patterns, which can naturally avoid chip errors caused by

¹Without loss of generosity, we use only 6 chips per symbol to illustrate the idea. In ZigBee, one byte has two symbols, each of which has 32 chips.

Cyclic Prefixing. In other words, the sequence “110011” can be better emulated by Wi-Fi OFDM under CP.

As a ZigBee preamble consists of a sequence of zero symbols followed by a SYNCWORD, LongBee first splits each zero symbol into left and right parts as shown in Fig. 11. Let 0_L be the left part and 0_R be the right part of the symbol 0. We choose the split position to minimize the CP-errors of $0_R||0_L$, then the SYNCWORD symbols are selected with minimum CP-errors using the exhaustive search, for the optimal split position only occurs at one of the borders of 32 chips constituting the ZigBee symbol 0. Many ZigBee receivers support the user defined SYNCWORD, such as CC2420 ZigBee chipset [12].

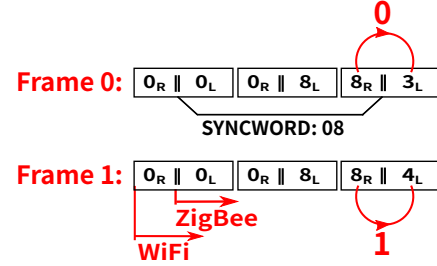


Fig. 11. Architecture of Transition Coding

We note that with the processing of SYNCWORD shown in Fig. 11, the ZigBee symbol boundary is no longer aligned with the Wi-Fi symbol boundary, such misalignment won't deteriorate the performance of ZigBee preamble detection, for the redundant symbols of the SYNCWORD will be ignored by the correlator as noises. However, such misalignment leads to transition coding to the payload in order to ensure correct demodulation of a whole ZigBee frame. Specifically, after preamble detection, LongBee encodes the bits of the ZigBee frame with the transition pattern. For example (Fig. 11), the transition from the right part of symbol 8 to the left part of symbol 3, denoted as $8_R||3_L$, can be coded as the bit “0”, while $8_R||4_L$ represents the bit “1”. We select ZigBee symbols 3 and 4 in a way that the CP-errors of $8_R||3_L$ and $8_R||4_L$ are lowest among all 16 possible transitions. LongBee encodes one bit when two transitions are used, and two bits when four transitions used, etc, and the data rate declines to 64KBps when two transitions used.

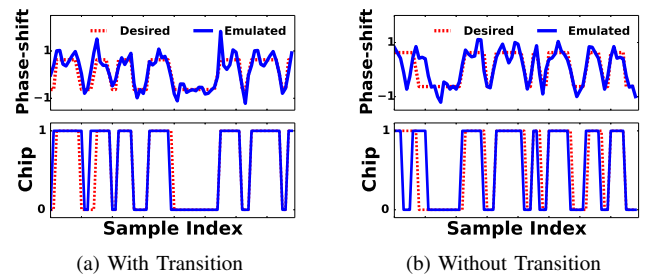


Fig. 12. Comparison of ZigBee Demodulation w/ and w/o Transition Coding

As a proof of concept example, the ZigBee symbol “0” is emulated with and without such transition coding mechanism. Fig. 12 illustrates the phase shifts and chips of the emulated

signals after transition coding. LongBee significantly decreases the amount of incorrect emulated chips with transition coding.

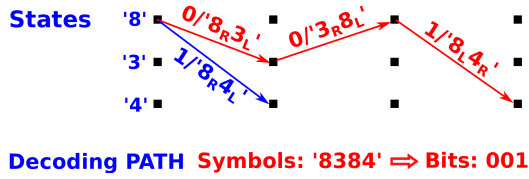


Fig. 13. Transition Decoding based on Viterbi Graph

C. Transition Decoding

LongBee encodes the bits of original ZigBee frame as the transitions of right and left parts of ZigBee symbols. When ZigBee device receives a sequence of ZigBee symbols, i.e., the state of Viterbi graph shown in Fig. 13, it also obtains a path composed of the transitions of ZigBee symbols. Based on this path, LongBee can compute a decoding path whose distance to the received path on the Viterbi graph is minimized and can decode the transmitted bits corresponding to the decoding path[23]. Because the sensitivity of both preamble detection and payload reception is improved with transition coding, the ZigBee receiver can receive emulated ZigBee from Wi-Fi over a significantly longer distance.

VI. PERFORMANCE EVALUATION

As shown in Figure 14, the sender of LongBee is implemented on a USRP-B210 platform with 802.11 b/g PHY[3]. For comparison, we implement two baselines: a ZigBee sender (0 dBm) on 802.15.4 PHY[19] and a WEBe sender (26 dBm) on 802.11 b/g PHY [3] in our testbed. To evaluate how accurately LongBee can emulate ZigBee signal, we also implement a boosted ZigBee transmitter with 26dBm Tx power. We note the boosted ZigBee can only be realized through USRP, serving as the Oracle upper bound for comparison. The receivers of LongBee and baselines are all implemented on a commodity ZigBee device (i.e., TelosB with CC2420 chip).

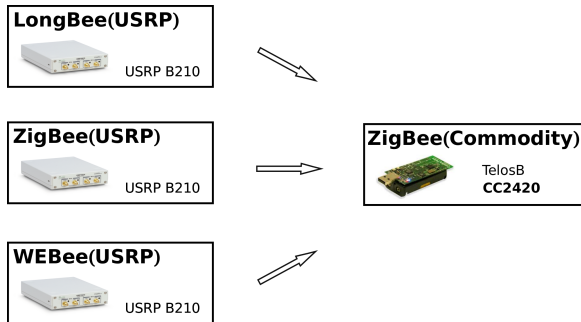


Fig. 14. Experiment Setting for LongBee

During experiments, each emulated frame consists of a 32-bit preamble, and a default frame payload of 25 bytes, resulting in about a 1ms duration, which is the typical length of payload for ZigBee communications. To ensure statistical validity, we obtain the average result of 10 experiments, each of which sends 1,000 LongBee/WEBe/ZigBee frames under a wide

range of settings including indoor/outdoor, mobile, and duty-cycled scenarios.

A. Experimental Results

The evaluation of LongBee mainly focuses on the link-layer statistics (i.e., frame reception ratio (FRR)) in long range communication. We conduct comprehensive experiments in multiple typical sites, including (i) an indoor hall, (ii) an indoor parking lot, and (iii) an outdoor plaza. Moreover, our experiments also cover multiple wireless scenarios, including mobile and low-duty-cycled environments.

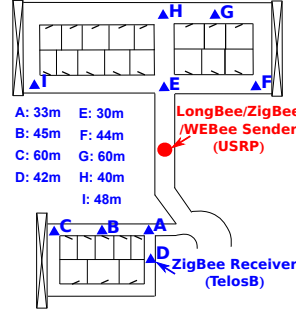


Fig. 15. Hall Site.

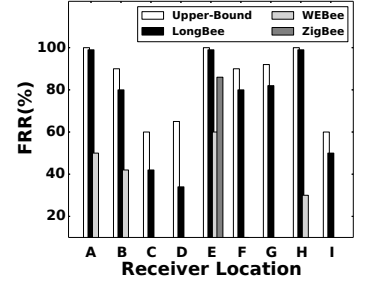


Fig. 16. Frame Reception Ratio at Different Locations (Hall).

Long Range Performance (Indoor): We first evaluate the long range performance of LongBee in two indoor sites: (i) a hall (Fig. 15) and an indoor parking lot (Fig. 17). We set the transmission power of LongBee, ZigBee, and WEBe at 26dBm (the maximum transmitter output power allowed by FCC under ISM band is 30dBm) and only change the transmission distances. Fig. 16 and Fig. 18 show those FRR between the LongBee/WEBe/ZigBee sender and the ZigBee receiver under varying distances.



Fig. 17. Indoor Parking Lot Site.

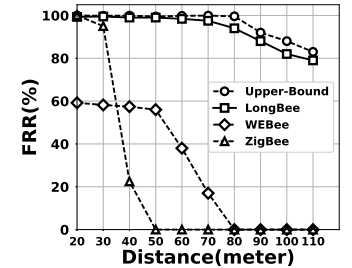


Fig. 18. Frame Reception Ratio with Tx Distances (Parking Lot).

In line-of-sight scenarios (Parking lot), LongBee delivers 90% frames, even the transmission distance is above 90 meters (Fig. 18), while the FRR of normal ZigBee drops to 90% with a distance of merely 35 meters. We note that LongBee has a very similar performance as high-power ZigBee (i.e., 26dBm), proving the effectiveness of signal emulation. Fig. 18 also shows that the WEBe's transmission distance is about 60% of LongBee, due to its less concentrated transmission power and low receiver sensitivity. For the indoor site with



Fig. 19. Outdoor Site.

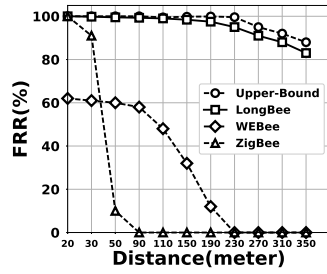


Fig. 20. Frame Reception Ratio with Tx Distances (Outdoor).

the complicated structure shown in Fig. 15, LongBee still has a comparable range performance with high-powered ZigBee.

Long Range Performance (Outdoor): We also evaluate the long range performance of LongBee in an outdoor site, i.e., a plaza shown in Fig. 19. We set the transmission power of LongBee, ZigBee, and WEBe at 26dBm and change the transmission distances. Fig. 20 illustrates that the FRR of LongBee is more than 90% even the transmission distance is above 300 meters, indicating that LongBee can effectively support long range communications for outdoor environments. Furthermore, evaluations at the outdoor site indicate that LongBee has a comparable performance with high powered ZigBee (26dBm). In contrast, the WEBe's transmission distance is only 160 meters, less than 50% of those of LongBee, and the communication distance of regular ZigBee (0dBm) is only about 30 meters, less than 10% of those of LongBee.

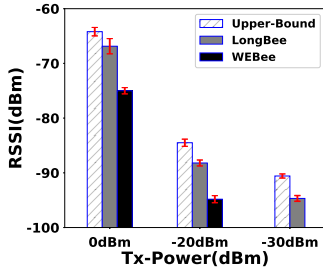


Fig. 21. RSSI with Tx Powers (Room).

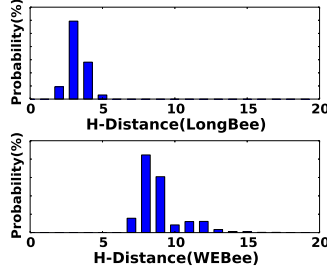


Fig. 22. H-Distance of LongBee vs. WEBe.

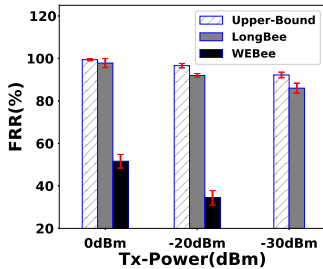


Fig. 23. Frame Reception Ratio with Tx Powers (Room).

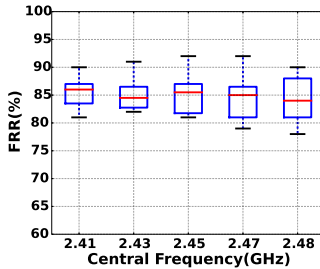


Fig. 24. FRR with Wavelength (Room and Tx-power = -30dBm).

Experimental Insights: To understand the superior performance of LongBee in comparison with WEBe and ZigBee, we conduct another set of experiments to obtain physical-level measurements. In a room site, we measure the Received Signal

Strength Indicator (RSSI) and the distribution of chip Hamming distances between emulated signals and intended ZigBee symbols for LongBee, ZigBee, and WEBe under different transmission powers, i.e., 0dBm, -20dBm and -30dBm. We note (i) RSSI at the receiver reflects the effective transmission power at the sender, and (ii) the Hamming distance caused by chip errors reflects the sensitivity of the receiver. Moreover, we also measure the FRR of LongBee, ZigBee and WEBe with same wireless configurations.

Fig. 21 proves that LongBee can increase the effective transmission power significantly. In comparison with ZigBee, LongBee only loses less than 4dBm Tx-Power while WEBe loses 12dBm. Consistent with RSSI, the FRR in Fig. 23 shows similar features, i.e., LongBee's performance approaches the upper bound performance (that of boosted ZigBee) even with very low transmission powers. In contrast, WEBe can't receive any frames when the Tx-Power is lower than -30dBm.

Furthermore, Fig. 22 shows that the Hamming distances of LongBee are mostly in the region of [2,5], much less than those of WEBe (i.e., [7,12]). As a result, the reliability of LongBee is much higher than WEBe (e.g., 99% to 50% with 0dBm Tx-power in Fig. 23). This result is a direct effect of transition coding in improving the receiver sensitivity.

We note that the communication distance also depends on the wavelength, i.e., the λ in equation (1). Fig. 24 evaluates the impact of the wavelength on the frame reception ratio. LongBee builds a communication from WiFi to ZigBee on the spectrum from 2.41GHz to 2.48GHz, so the wavelength varies from 12.09cm to 12.44cm. Experimental results show that the impact of wavelength is ignorable for LongBee, for the difference in wavelength only leads to 0.25dB variance in receive signal power.

Performance under Mobility: We also evaluate LongBee with a mobile ZigBee receiver. We conduct this experiments in the outdoor site (Fig. 19) and set the transmission distance as 200 meters. Six levels of movements are evaluated in this experiment, namely, strolling, walking, striding, jogging, running and biking. Fig. 25 shows that LongBee works well under different levels of mobilities.

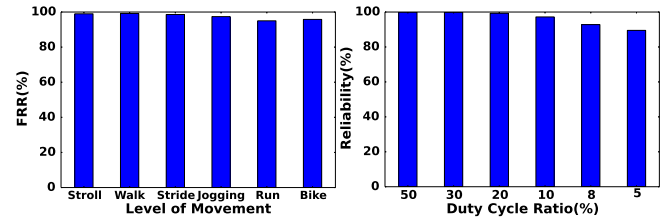


Fig. 25. Frame Reception under Mobility.

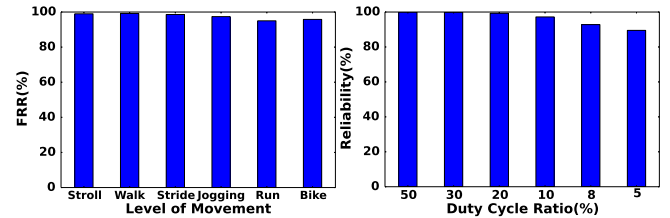


Fig. 26. Frame Reception Ratio under Low Duty-Cycle.

Performance under Low Duty-Cycles: LongBee is also evaluated in the low-duty-cycle scenarios, in which ZigBee receivers sleep most of the time and wake up periodically to check for radio activity. In this case, a Wi-Fi sender needs to transmit a train of repeated frames to wake up the receiver [4]. We evaluate LongBee in ContikiOS [4], and each LongBee

frame is retransmitted 40 times. The duty-cycle ratios are set as 50%, 30%, 20%, 10%, 8% and 5%, respectively. The experiments are conducted in the outdoor site (Fig. 19) with 200m transmission distance. Figure 26 shows that the LongBee can work well under a wide range of duty-cycle settings.

VII. RELATED WORKS

Cross-technology communications (CTC) has emerged as a promising mechanism for explicit collaboration between heterogeneous wireless communications. FreeBee [14] establishes CTC by modulating the interval of Wi-Fi Beacons. B2W2 [5] achieves the CTC by manipulating the transmission power of BLE packets, etc. However, these packet-level CTC [14], [5] can't support long range wireless communication because the RSSI required by the receiver is very high (e.g., -75dBm in FreeBee[14]) to support reliable energy detection which is the foundation of packet-level CTC.

In comparison with packet-level CTC, WEBee[15] is a pioneering work to implement the CTC via physical signal emulation, where Wi-Fi OFDM generates ZigBee physical signals by manipulating the Wi-Fi payload. Similar to WEBee, LongBee uses physical-level emulation for communication. Departure from WEBee, LongBee uniquely extends its communication range by concentrating the effective TX power through down-clocking and improving the RX sensitivity using an innovative transition coding. Although the description is specific, these designs can be generically applied to other wireless communication.

Existing researches on enabling long-range communication in wireless networks (i.e., TinyNode[7], Fleck-3[6], and SNOW[18], LoRa [10], and NB-IOT [17]) mostly relied on hardware modification such as increasing the transmission power, adjusting the channel coding or even implementing the new modulation. Departure from the literature, the unique features (i.e., asymmetry bandwidth and emulation error) in CTC inspires the innovative design of LongBee, which, to our best knowledge, has never proposed in the literature and are complementary to the other works [7], [6], [18], [10], [17].

VIII. CONCLUSION

Cross technology communication is a promising direction for wireless channel coordination and collaboration by allowing heterogeneous devices to exchange data directly without gateways. This work presents LongBee, the first work that focuses on extending communication range of cross technology communication. It has two innovative designs: (i) down-clocked emulation with pilot/null subcarrier handling, and (ii) transition coding for reliable preamble detection. Our extensive experiments show that LongBee extends communication range more than 2x those of the current state of art CTC with more than 99% frame reliability, and about 10x those of low power wireless communication in IoT (i.e., ZigBee with 0dBm transmission power). We expect that LongBee can amplify the applicability of CTC significantly due to its long communication distance, e.g., build low-power wide area (LPWA) applications with the help of existing Wi-Fi infrastructures.

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